

Development of a Blockchain-Based Smart City Monitoring Dashboard Using IoT Sensors and Ethereum Smart Contracts

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Abstract—Smart city technologies require reliable systems capable of collecting, storing, visualizing, and protecting urban monitoring data. This paper presents a blockchain-based smart city monitoring dashboard that integrates IoT devices, database systems, web visualization technologies, and Ethereum blockchain mechanisms. Environmental information is collected through an ESP8266 NodeMCU and DHT11 sensor, while additional smart city indicators including traffic density, air quality, electricity consumption, water levels, and security alerts are simulated for demonstration purposes. The backend system is implemented using Python Flask and SQLite. To ensure data integrity, SHA-256 hashing and Ethereum smart contracts deployed on a local Ganache blockchain are utilized. Experimental results demonstrate that the proposed system successfully performs real-time monitoring, historical data management, blockchain verification, and secure transaction recording.

Index Terms—Blockchain, Smart City, Ethereum, Ganache, IoT, ESP8266, Flask, Dashboard, Smart Contract, Data Integrity

I. INTRODUCTION

The rapid development of urban areas has significantly increased the complexity of city management. Modern cities must continuously monitor transportation systems, environmental conditions, public utilities, water resources, and security infrastructures. Traditional monitoring approaches often rely on isolated systems that operate independently, resulting in fragmented information and delayed decision-making processes.

The concept of Smart City has emerged as a solution to these challenges by integrating sensors, communication technologies, cloud services, and intelligent management platforms. Through real-time data acquisition and centralized monitoring, city administrators can obtain a comprehensive view of urban operations and respond more effectively to potential incidents.

Despite these advantages, one major concern remains the integrity and reliability of collected monitoring data. Since smart city decisions are based on sensor-generated information, unauthorized modification of data may lead to incorrect actions and operational risks.

To address this issue, blockchain technology can be integrated into smart city platforms. Blockchain provides immutable record storage through cryptographic hashing and distributed verification mechanisms. By combining IoT technologies with blockchain verification, monitoring data can become more secure and trustworthy.

This project proposes a Smart City Monitoring Dashboard that combines IoT sensing, centralized visualization, historical data storage, and blockchain-based integrity verification within a single platform.

II. RELATED WORK

Numerous studies have investigated the integration of IoT and blockchain technologies within smart city environments.

Traditional smart city systems primarily focus on data collection and visualization. These systems typically utilize sensor networks and cloud infrastructures to provide real-time information regarding traffic conditions, environmental quality, and energy consumption.

Recent research has explored blockchain-based approaches to improve data security. Blockchain enables tamper-resistant storage through cryptographic mechanisms and distributed consensus algorithms. Ethereum has become one of the most widely adopted blockchain platforms due to its support for smart contracts and decentralized applications.

Several blockchain-enabled smart city frameworks have been proposed. However, many existing solutions require complex infrastructure and significant computational resources. Therefore, this project adopts a lightweight architecture suitable for educational and demonstration purposes while preserving the essential principles of blockchain verification.

III. SYSTEM ARCHITECTURE

The proposed system consists of five primary layers:

- 1) IoT Sensor Layer
- 2) Backend Processing Layer
- 3) Database Layer
- 4) Blockchain Layer
- 5) Dashboard Visualization Layer

The overall workflow is illustrated in Figure 1. The IoT layer is responsible for collecting environmental information through physical sensors. The backend server receives incoming data and processes monitoring records. The database stores historical information for later retrieval. The blockchain layer generates cryptographic hashes and records transaction information on Ethereum. Finally, the dashboard layer presents data through charts, maps, and monitoring panels.

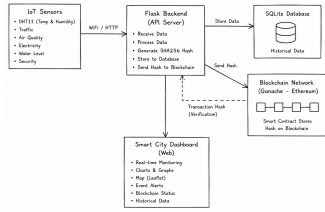


Fig. 1. Overall system architecture

IV. IOT DATA COLLECTION LAYER

The hardware component of the proposed system utilizes an ESP8266 NodeMCU development board together with a DHT11 temperature and humidity sensor.

The DHT11 continuously measures environmental conditions. Sensor readings are transmitted to the ESP8266 through digital communication pins. The ESP8266 processes collected values and forwards them to the backend server via Wi-Fi communication.

The measured parameters include:

- Temperature (°C)
- Humidity (%)

For demonstration purposes, additional smart city indicators are simulated:

- Traffic Density
- Air Quality Index
- Electricity Consumption
- Water Level Monitoring
- Security Alert Status

This hybrid approach allows the system to represent a realistic smart city environment without requiring a large number of physical sensors.

V. BACKEND AND DATABASE DESIGN

The backend is implemented using the Flask framework in Python.

The system exposes several RESTful APIs:

- /data
- /history
- /verify

The /data endpoint collects sensor information and simulated city indicators.

The /history endpoint retrieves previously stored records.

The /verify endpoint validates blockchain consistency.

SQLite is selected as the database management system due to its lightweight architecture and ease of integration.

Each monitoring record contains:

- Record ID
- Traffic Information
- Air Quality Index
- Electricity Consumption
- Water Level
- Security Status

- Timestamp
- Previous Hash
- Current Hash

This structure supports both historical analysis and blockchain verification.

VI. BLOCKCHAIN-BASED DATA VERIFICATION

Blockchain technology forms the core security component of the proposed system.

For every monitoring record, a cryptographic hash is generated using the SHA-256 algorithm.

The hash generation process can be represented as:

$$[\text{Hash}_i = \text{SHA256}(\text{Data}_i + \text{PreviousHash}_i)]$$

where:

- (Data_i) represents monitoring data.
- (PreviousHash_i) represents the previous block hash.

Each newly generated block contains:

- 1) Monitoring Data
- 2) Previous Hash
- 3) Current Hash
- 4) Timestamp

The blockchain structure creates a chain of linked records:

$$[\text{Block}_1 \rightarrow \text{Block}_2 \rightarrow \text{Block}_3 \rightarrow \text{Block}_n]$$

If any previously stored data are modified, the corresponding hash changes immediately, causing a mismatch in subsequent blocks.

The verification algorithm compares each block's Previous Hash with the hash of the preceding block. If inconsistencies are detected, the system reports a blockchain integrity violation.

This mechanism enables rapid detection of unauthorized modifications.

VII. ETHEREUM SMART CONTRACT INTEGRATION

In addition to local blockchain verification, Ethereum smart contracts are integrated into the system.

The smart contract is written using Solidity and deployed to a local Ganache blockchain network.

The contract stores monitoring hashes generated by the backend system.

A simplified contract structure is shown below:

```

contract SmartCity {

    string public latestHash;

    function storeHash(
        string memory _hash
    ) public {

        latestHash = _hash;
    }

}

```

Whenever a new monitoring record is created, the backend submits the generated hash to the smart contract.

Future work will focus on cloud deployment, additional sensor integration, machine learning analytics, and public blockchain deployment.

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